

Scheduling Tasks in Linux



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Objectives



Understanding Task Scheduling

- systemd-timer units
- cron
- at





Task Scheduling

Traditionally Unix and Linux have used **crond** as the task scheduler. We also have the **atd** and **systemd** timer units



Task Scheduling

crond

**Jobs at regular
intervals**

atd

**Once off, or
irregular tasks**

timer units

**No need for
additional service**



```
$ man 5 systemd.timer
```

Documentation

Being able to quickly access help documentation will be an asset in the exam and in real life




```
[Unit]
Description="Back up the etc directory"

[Service]
ExecStart=tar -czf /root/etc.tgz /etc
```

Service Unit

The timer unit will run a service unit. The service unit will not have the [Install] section meaning that it cannot be enabled.



```
[Unit]
Description="Run backup of etc once a day"

[Timer]
OnBootSec=5min
OnUnitActiveSec=24h
Unit=backup.service

[Install]
WantedBy=multi-user.target
```

Timer Unit

The timer unit calls the service. We enable the timer unit, so it starts automatically. In this configuration it will run 5 minutes after boot and the every day.



```
$ sudo systemd-analyze verify /etc/systemd/system/backup.service
$ sudo systemd-analyze verify /etc/systemd/system/backup.timer
$ sudo systemctl enable backup.timer
$ sudo reboot
$ systemctl list-timers
```

Verify Units

The analyze command that we saw previously can also be used to verify unit files



Demo



Working on Ubuntu1

- investigate system timer units
- create a new service unit
- and trigger it with a timer



```
$ grep '^#' /etc/crontab  
$ ls /etc/cron.d/  
$ sudo nano /etc/cron.d/test  
*/10 * * * * root date >> /tmp/date
```

System Crontab

The root user can maintain scheduled jobs with the `/etc/crontab` file but more likely, `/etc/cron.d/`. In the sample cron entry we create, every 10 minutes we write the date to a file.



```
$ crontab -l  
$ crontab -e  
$ crontab -r
```

User Crons

Users can maintain their own crontab file using the crontab command. The fields are the same as the system cron file but does not include the user account under which to execute.



Demo



Managing Cron Jobs

- system cron jobs
- user crons



```
$ sudo systemctl status atd  
$ atq  
$ at noon  
$ at 10:51 July 25 2023  
$ atq  
$ atrm 2  
$ at -c 1
```

Once Only Tasks

The atd can be used to scheduled tasks that are not needed on a regular basis



Demo



Working with At

- We now look at the **atd**



Git Repo

```
git clone https://github.com/theurbanpenguin/lfcs
```



Summary



Scheduling Tasks in Linux

- systemd timers
- service unit
- timer unit
- systemd-analyze verify
- /etc/crontab /etc/cron.d
- crontab
- atd



Up Next:
Monitoring Processes in Linux

